12.7 Modeling with Surface Area and Volume

Name _____ Period ____ Date _____

Geometry 12.7 · Student sheet

I can…

- find and use density.
- compare the surface areas and volumes of similar solids.
- find the box with the least surface area for a given volume.

Vocabulary · fill in the missing word

Density — the _____ of matter that an object has in a given unit of volume: mass ÷ volume

Similar solids — two solids of the same type with equal ratios of _____ linear measures

Part 1 · Discovery: same metal, different cubes

Three aluminum cubes. Fill in the table.

Edge	Mass	Volume	Surface area	Mass ÷ volume
1 cm	2.7 g			
2 cm	21.6 g			
3 cm	72.9 g			

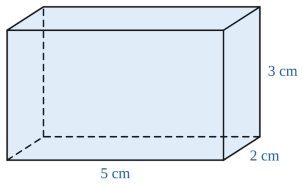
When the edge is multiplied by 2, the surface area is multiplied by _____ and the volume by _____.

Part 2 · Notes and examples (fill in each note, then work the example)

Density = _____ ÷ _____

Example 1

A metal block is 2 cm by 3 cm by 5 cm and has a mass of 237 grams. Find its density.



Example 2

A copper cylinder has a radius of 1 cm and a height of 5 cm. Copper's density is 8.96 $\frac{g}{cm^3}$. Find the mass of the cylinder. Round to the nearest hundredth.

Similar solids: with scale factor k, the surface areas are in the ratio _____ and the volumes are in the ratio _____.

Example 3

Two similar prisms have a scale factor of 3 : 5. The smaller has a surface area of 90 ft² and a volume of 54 ft³. Find the surface area and volume of the larger.

Example 4

Two similar cylinders have volumes of 64π and 216π. Find the scale factor and the ratio of their surface areas.

Least surface area: for a given volume, the box with the least surface area is a _____.

Example 5

Which box with a volume of 64 in.³ has the least surface area: 4 by 4 by 4, 2 by 4 by 8, or 1 by 8 by 8?

Part 3 · Partner practice

Partner 1

A block has a volume of 125 cm³ and a mass of 340 g. Find its density.

Partner 2

Two similar cones have a scale factor of 1 : 4. The smaller has a volume of 12 cm³. Find the volume of the larger.